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Multi-user Collaborative Volume Data Illustration and Visualization

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Summary

Multi-user can collaboratively complete complex visualization tasks that cannot be completed by a single user. Although multi-user collaboration system has made great progress, there are many challenges in the collaborative visualization of 3-D volume data due to the difficulties in multi-user collaboration, and collaborative slice analysis. This paper proposes a client-server (CS)-based collaborative visualization system, which consists of a 3-D volume explorer and a 2-D slice analyzer, to help domain experts to fully utilize their background domain knowledge to illustrate different parts of the data. For example, the brain surgeon expert, pulmonologist, and cardiologist can visualize and analyze the different subsets of the volume data, i.e., the corresponding sub-volumes of the brain, heart, lungs, and blood vessels. It also allows to take full advantage of the hardware resources, because all the computation intensive tasks especially for the whole data rendering can be allocated to the powerful sever while the light-weight tasks can be allocated to the portable clients. We evaluate the system by collecting the feedback from domain experts and the people who are unfamiliar with data computation or visualization. The domain experts' feedback shows that the 3-D volume explorer and the 2-D slice analyzer are capable of supporting peer-expert discussion and medical case teaching, respectively. Furthermore, the evaluation shows the usability of the system for the people who are unfamiliar with the domain fields.

KEYWORDS:
Scientific data visualization, Collaborative visualization, Visualization software

1 | INTRODUCTION

Most of the current scientific visualization systems usually allow only one user to process and analyze the data in a single machine. The single-user single-machine systems show some limitations for complex data analysis in visualization. First, it is inefficient when a team of experts want to illustrate the different parts of the data simultaneously or share their illustrations [1,2]. Second, all the tasks of computation, rendering and interactions are assigned onto one machine, which would cause some performance issues especially when the data are too large to handle in a commonly-used computer [3]. Third, it is impossible for the experts in different domain fields to utilize their background knowledge in data illustration and visualizations [2,4,5]. Multi-user and multi-machine collaborative visualization systems could solve the above-mentioned three issues significantly.

Early works about collaboration mostly focus on co-located visualization and 2-D data visual analysis, which base on the low rendering power devices, such as web browser and VR glasses. The co-located visualization system is mainly implemented by web technology, which has good compatibility and portability. However, the performance of web-based approaches is worse and the affordable data size is limited, which is often not available in 3-D rendering. With the analysis of 2-D data is mainly used in the low ability machine, which has good real-time interactivity. In the face of complex 3-D data visualization, the existing systems will be limited.

Collaborative visualization and analysis relying on network allows multiple users to observe and interact with the same data together[6]. Such a system can perform tasks that cannot be completed by a single person, or allow users to discuss with a collaborator to explain and confirm the analysis results. Although most visualization techniques are developed for a single user[7], many early studies have successfully implemented multi-user visualization[8,9]. Effective collaboration technology involves design, communication, joint analysis and decision-making, which can effectively accelerate the analysis of 3-D data.

The data analysis is usually a subsequent task for collaborative analysis[10]. Multi-user collaboration can be a great boost for illustration and visualization process. Previous efforts mainly focus on how to improve the rendering effect. Such as a better transfer function assigning scheme for volume rendering[11,12,13]. Now, more new interactive issues need studying, including multi-user collaborative work[14], coordination and communication. In addition, multi-user interactive technology still needs future exploration.

The analysis of 3-D data can better display the overall structure of the data, and 2-D data can better display the detailed features in the data. In the 3-D volume data analysis, interactive direct volume rendering is one of the most effective methods to visualize and reveal the significant features presented, and are well applied in medical application[15] or geological field[16]. It enables domain experts to intuitively discover scientifically meaningful features and have a better spatial perception of the data. Besides, 2-D data have some certain advantages when presenting details of certain parts. So slice interpretation is an important way that extract local 2-D data from original data sets and present it distinctly for more detailed analysis. By interpreting and editing the slice, more slight details will be presented clearly. It can help domain experts focus on a certain piece of data, thus having less distraction from unrelated parts. With volume analysis and slice analysis combined, domain experts are able to view the data globally to grasp general spatial features, and then extract certain parts and interpret it for more thoroughly study. Sometimes experts can inserted the interpreted slice back to the volume rendering result for comparison, making the interpretation process efficient and effective.

In this work, we propose a collaborative and interactive visualization system. It consists of a 3-D volume data explorer and a 2-D slice analyzer. The system adopts a client-server framework and it allows to assign the computation-intensive tasks to the server and the light-weight tasks to the client. The server equipped with more powerful computing resources could be used to render the whole 3-D volume data, while the client could be used to process some light-weight tasks, e.g., 2-D slice analysis, sub-volume rendering and other interactions, which often are some portable interactive devices. This system enables users with different background expertise to illustrate the corresponding sub-volumes cut and registered by them, and make real-time adjustments with the help of transfer function editor. For example, in medical data visualization, the brain surgeon, pulmonologist and cardiologist can work together to illustrate the corresponding tissues (sub-volumes) by using the proposed approach. For the 2-D slice analyzer, users can freely select slices in a certain dimension and perform 2-D interactions with the help of illustrating tools such as pencil, eraser, circle selection, lasso selection, 2-D transfer function editor, etc. Finally, the edited volume results and the illustrated slices can be fused into a single 3-D environment to provide a focus+context exploration scheme for users.

Then, we conducted some user studies with participants without medical expertise to evaluate the collaborative experience of multi-user and multi-machine system. The studies in a real personal context demonstrated that the participants were able to render the 3-D volume data and select 2-D slice data in a real-time interaction, and were satisfied with the system subjectively. Additionally, we invited some medical experts to try out our systems, and received some appreciations and helpful proposals for the system. In addition to the medical data visualization, the multi-user and multi-machine collaborative visualization systems could also be used for the petroleum data and seismic data visualization, which contributes the related domain experts to illustrate and adjust the corresponding sub-volumes cut in real time.

The rest of the paper is organized as follows. In Section 2, we surveyed the related work. We describe the methodology of the collaborative visualization system in Section 3. Then we describe the evaluation including several case studies and user studies, and domain experts' feedback in Section 4. The limitations of the proposed approach are discussed in Section 5. Finally, we summarize the work and discuss some potential future extensions in Section 6.

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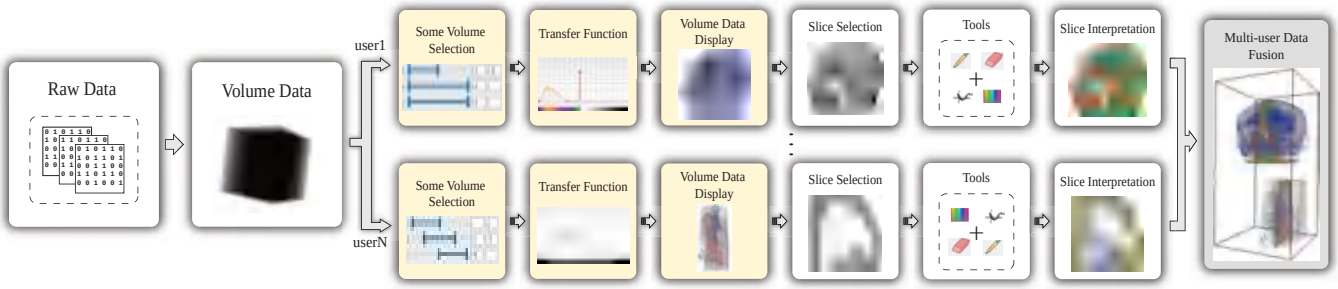

FIGURE 1 Pipeline of the multi-user collaboration. Raw data is reprocessed into a volume. The user applies for the scope of the editing area and the position of the slice. All users edit and draw the applied data with the help of system tools. All the slice data and volume data edited by the users are merged into the final interpretation.

2 | RELATED WORK

The related work can be roughly categorized into collaborative visualization, interactive visualization and volume cut visualization.

2.1 | Collaborative Visualization

Early works about collaboration mostly focus on co-located visualization, which only need to deal with multi-input and visualization effects[17,18,19]. However, recent years have seen great advance in web technology, making remote collaborative systems based on web more practical.

Some systems use web-based technologies to create data-sharing platforms, which can be accessed by every portable terminal with browsers in it[20,21]. For example, Puel et al proposed a web-based solution for the remote medical image visualization problem, and integrate both visualization and communication on a single cooperative platform[22]. However, despite good portability, those systems have limited capacity visualizing complex data (only PNG format is supported in Puel's work).

Some other works provides immersive experience by creating 3-D collaborative virtual environment (CVE) for multiple users to explore together. For example, Mahmood et al. provided an implementation features of co-presence, remote information sharing, and co-analysis in his collaborative visualization system based on mixed reality technics[6]. Chheang et al. presented a VR system that enables doctors to cooperate and simulate laparoscopic liver surgery[14]. Lavric et al. developed a Med3D medical data visualization framework with camera parameters attached to annotations, making view angle shareable[23]. Alma et al. proposed a synchronize algorithm with small delay, making each user updated in a 3-D CVE[24].

However, only a few of the existing visulization systems provide practical method to visualize volume data and interactively manipulate slices for remote collaborative teams. Because large dataset and complex visulization algorithm are difficult challenges for an online system, especially when the client has poor computing performance, or unstable web connection, researchers need to consider and optimize data transfer and visualization method.[25,26,27,28]. In this field, Wu et al. proposed a medical imaging system that support slice rendering and volume rendering, and make up for the deficiency of cooperation and cross-platform access in traditional medical visualization systems[29,30]. Deng et al. presented an online 3-D image rendering platform, sharing camera angles in real-time[31]. These systems have implmented volume rendering and collaboration in real-time, some can change color schemes of different data parts, allowing flexibility and customization in interaction. But they still have poor ability when division of labor is needed. Doctors with different speciality actually can work on the same piece of data without distraction from others, yet the result can still be shown on the whole in real-time.

2.2 | Interactive Volume Visualization

Volume visualization can display 3-D data sets from 2-D image slices.[32]. It is usually applied to volume slices of medical radiology data, or 3-D data sets in the oil and gas industry[33]. One of the most commonly used visualization algorithms is ray-casting algorithm. It emit a light from each pixel, traverses the entire image sequence and sample the color information to

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compute the final color value[34,35]. This method has been widely applied in medical data sets such as CT and MRI[15,36,37] and seismic cube data[16].

However, due to the high computational cost of volume rendering, most of the volume visualization system need to run on high-end graphics cards, and may still has low refresh rate with complex algorithm such as ray tracing[3]. Further more, it's even harder to provide low-latency interaction when dealing with large data sets. As a result, many systems allow users to manipulate data with multi-resolution on multi-scale[38,4], These approaches can significantly reduce computational cost and memory cost. Some other systems use advanced algorithms to improve performance for specific datasets, such as neuroscience data[39] or large meshes[40]. Further more, some systems attempt to achieve interaction on different platform using WebGL[41], or client-server architectures[5]. These systems either require small amount of computational resources, or avoid complex configuration, making volume rendering in different usage scenarios possible.

Due to the complexity of volume data set, it's important for users to interactively extract meaningful information from the original data[39]. With respect to this, many interactive tools are proposed in certain fields. For example, for neurosurgery planning, Diepenbrock et al. designed a toolbox to visualize data comprehensibly[42]. Also, in the field of seismic analysis, horizons visualization[43,44] and faults visualization [45,46,47] techniques makes it easier to interpret the noisy, dense and ambiguous seismic data. Liu et al. proposed some methods that focus on specific stratigraphic structure extraction and visualization. They use the transfer function to first edit the seismic data and create professional slices, and then allow experts to analyze the slices with a slice analysis system[2,1]. To realize remote collaboration, our approach implements a network version of transfer function for volume visualization and slices interpretation. This approach is also based on client-server architecture, achieving a balance between interactive visualization and efficient collaboration.

2.3 | Sketch-based Visualization

The sketch-based interpretation visualization method is an integrated method that combines human factors and special knowledge. It can optimize the traditional visualization methods, namely WIMP(window, icon, menu, pointer) paradigm, so that even novices who have insufficient professional knowledge and programming skills can quickly interpret professional data[48].

One of the main challenges of sketch-based visualization is sketches interpretation. There are some common methods for sketches interpretation, such as color mapping, segmentation of different regions, and geometric division of region[48]. For some noisy and discontinuous data sets, sketch-based visualization method can reduce the noisiness of the data set, allowing the data to be interpreted more intuitively. As a result, sketches are widely used in illustrative visualization systems for various data types. For example, Schroeder et al. designed a drawing interface which allows artists to interpret 2-D vector field data. Wei designed a sketch-based 3-D vector field classification interface, allowing the user to draw 2-D sketch curves to automatically match all similar 3-D streamlines[49]. Zhu proposed a sketch-based system to simulate internal flow pattern by gradually editing the fluid data[50]. Moreover, Guchev[51], Liu[2], Natali[52,53] designed a set of sketch-based drawing tools based on texture replacement. these tools allow users to use a pen as a sketch tool to quickly edit the data set.

Those sketch-based systems all provide simple interactive method to improve interpretion efficiency when dealing with complex data sets. Our system also applies sketch-based interpretation methods to the editing of multi-user volume data sets to simplify the slice interpretation operation and produce better visualization effects.

3 | OUR METHOD

In this paper, we propose a multi-user collaborative system that allows experts in different fields to explore and interpret different parts of the volume data simultaneously. Each expert can focus on different areas according to their interests and background knowledge.

We employ a client-server framework to provide computations (on server), collaborations (on server and client) and interactions (on client). The 3-D volume explorer rendering the whole volume data is mainly run on the server, while the 2-D slice analyzer, sub-volume renderer and interaction processing are on the client. In our system, domain experts, e.g., doctors from different clinical departments, or geographers and biologists in different departments are allowed to utilize their own background domain knowledge in revealing significant features and share their discoveries in an efficient and effective way, which is a great improvement for medical or geological peer-discussion and a large boost for the efficiency in data illustration.

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Figure 1 shows the pipeline of multi-user collaborative volume data illustration and visualization. First, we preprocess the raw data into volume data. Second, each user (a client) connects the server of the system and applies for the boundaries of sub-volumes he/she want to edit. Third, each expert interactively edits and illustrates the corresponding sub-volume data and explore the whole volume visualization result received from the server in real-time. The whole volume visualization received from the server. For 3-D volume data analysis, we provide several cooperative interactions: basic explorations, cube selection, and local transfer function editing for each sub-volume. Moreover, users can use system tools to choose the 2-D slice data from volume data for more detailed analysis. For slicing analysis, we provide a convenient toolbox including pencil, rubber, local transfer function and other tools. Finally, the slicing results from different experts are uploaded to merge with the volume visualization result, thus generating a final integral illustration.

3.1 | Collaborative Volume Explorer

In our system, users can use two selection tools to select certain parts of the original volume data, namely cube selection and ellipsoid selection. In cube selection mode, for example, users can edit those values to apply for a specific cube, while in ellipsoid selection mode, the applied volume are the ellipsoid that is tangent to a selected cube.

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Algorithm 1 `Multi-user_Collaboration()` function.

```
function MULTI_USER_COOLLABORATION(user_list, texture_list, volume_texture)
    while t ≤ directionLength do
        position = entryPoint + t×direction
        lookup = texture(volume_texture, position)
        // Traverse all users
        for u in user_list do
            if mode(u)==ellipsoid then
                T^2 = (position.x)^2/(u.a)^2
                +(position.y)^2/(u.b)^2
                +(position.z)^2/(u.c)^2
                β_(u) = step(T^2,1)
            end if
            if mode(u)==cuboid then
                if position ∈ cuboid(u) then
                    β_(u) = 1
                else
                    β_(u) = 0
                end if
            end if
            src = texture(texture_list(u), lookup)
            destColor.rgb = destColor.rgb
            + β_(u)×(1.0-destColor.a)
            ×src.rgb×src.a
            destColor.a = destColor.a
            + β_(u)×(1.0-destColor.a)×src.a
        end for
        t = t + step_size
    end while
end function
```

In terms of volume visualization, we apply different algorithm for client and server. For the client, we use traditional ray-casting algorithm for fast real-time rendering. The algorithm simulates the process of ray traversing the entire image sequence. During one ray integration, the traditional method samples all of the passing image sequence, accumulates color information,

and then calculates the final color of the rendered image, as shown in Figure 2a. However, different parts of the volume data may be edited by different transfer functions to achieve multi-user collaboration on the server. So we propose a modified ray-casting algorithm on the basis of the traditional one for the server. We divide each ray into multiple segments, as show in algorithm 1. In each segment, the sample points should be within the sub-volume boundary of an individual user, then we get a mapping between the points and their corresponding transfer function according to the boundaries. As a result, the server part can integrates all users' interaction to visualize the general data, as shown in Figure 3.

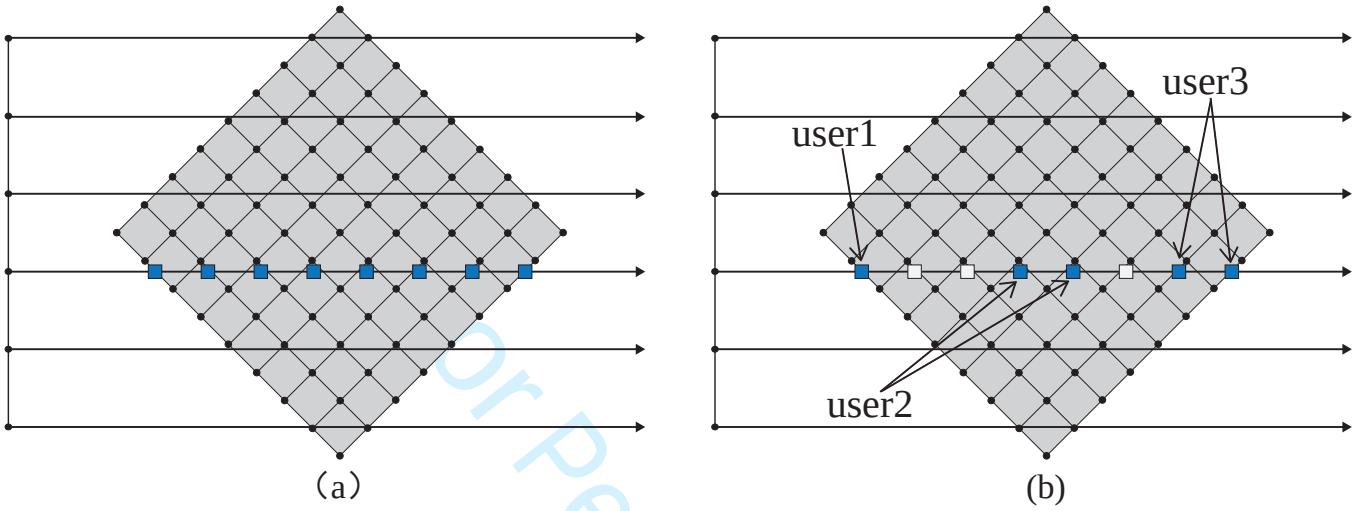

FIGURE 2 Ray-Casting algorithm. Left picture is the traditional continuous integration method[35]. Right picture is our discontinuous integration method. Different sample points falls into different users' edit area

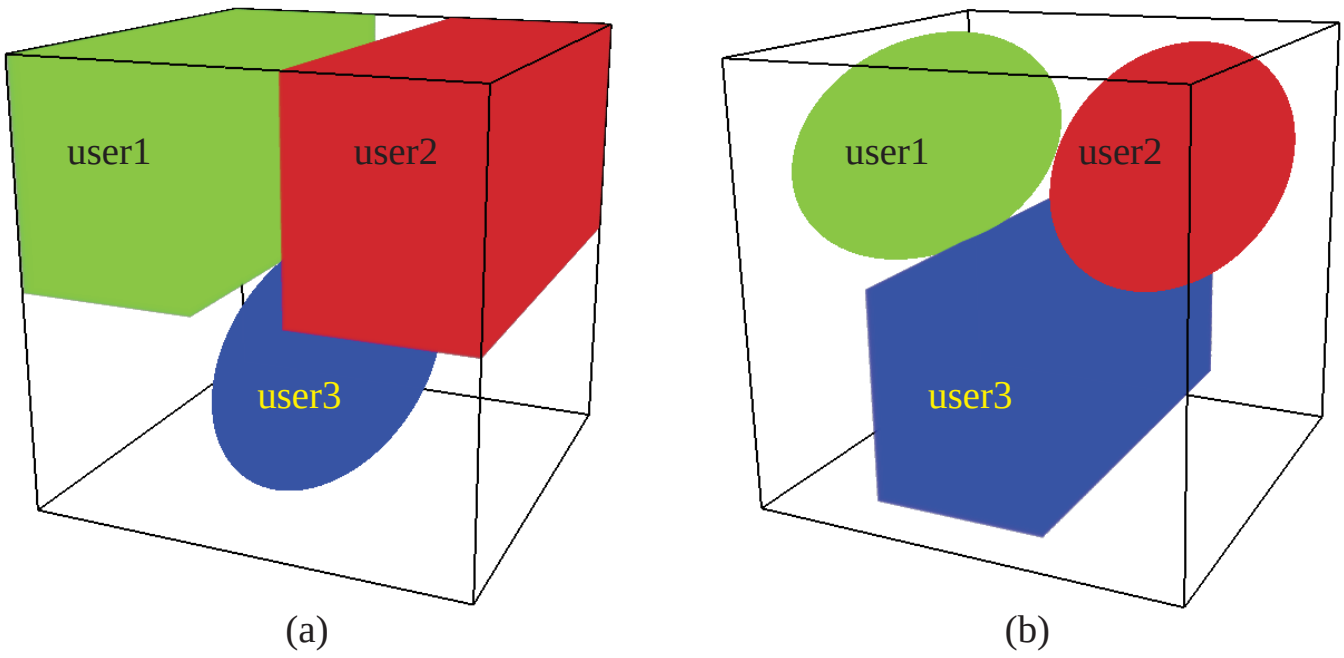

FIGURE 3 Region segmentation. Each user has his/her own edit area, which is indicated by different colors. (a) User1 and user2 use cuboids to identify their own areas, and user3 uses a closed surfaces to identify his/her ellipsoid area. (b) User1 and user2 use a closed surfaces respectively to identify their own ellipsoid area, while User3 uses a cuboid to identify his/her area.

The discontinuous integration method is the key to achieving multi-user collaborative interactions. Specifically, in server's each ray integration, we use a conditional integration method for the ray that penetrates the 3-D data as shown in Figure 2b. We use $\beta_{(u)}$ to determine whether the current sampling point meets the integration conditions. The method can be computed as follows:

$$C_{C_{(u)}} = \sum_{i=1}^{n} \int C_{(u+1)} \beta_{(u)} (1 - \alpha_{C_{(u)}}) du \tag{1}$$

$C_{(u)}$ and $\alpha_{C_{(u)}}$ represent the color value and opacity obtained by sampling on the volume texture. There are two cases for $\beta_{(u)}$: if it satisfies the boundary function specified by the current user, the value of $\beta_{(u)}$ would be 1, otherwise it would be 0. Finally, all segments are integrated into one ray. As the server machine is often equipped with better hardware resources and could get a better computation and rendering performance. All the computation-intensive and rendering-intensive tasks are assigned to the GPU on the server.

3.2 | Collaborative Slice Analyzer

3-D volume explorer can present the overall structure that is not visible in 2-D, but 2-D slice analyzer has unique advantages in presenting details for more thorough study. We allow different experts to edit and illustrate the sub-volume data, then illustrate the slices selected by them. The dual volume-slice illustration process allows experts to explore data from overview to detail and in a focus+context way. Users can select slices by choosing the corresponding positions from the volume data, and then edit them with 2-D slice analyzer.

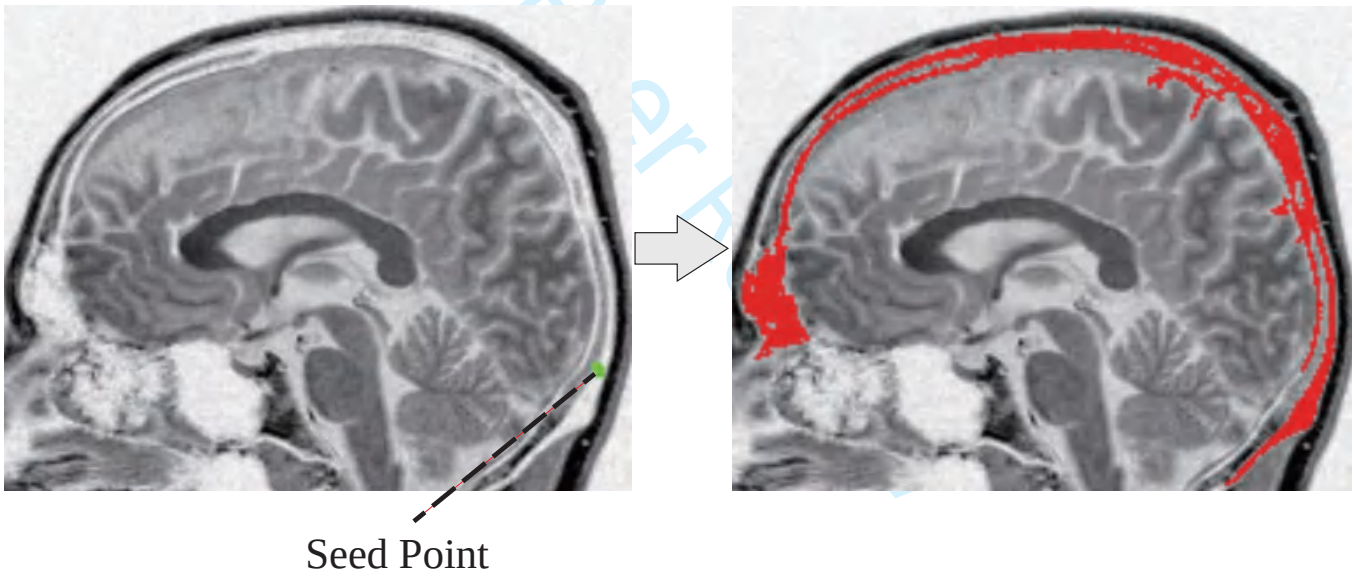

FIGURE 4 Automatic identification based on seed points. Similar areas are automatically identified and colored by seed points.

Several selection options are provided to assist to select and edit slices. Basically, users can draw a closed curve like lasso for further illustration. To provide more flexibility, we implemented a similar seed point search for area drawing, which is inspired by Liu et al[54]. We use flood-fill algorithm to quickly identify similarities in the area in the area drawing, as show in algorithm 2. Users can use mouse picking to determine seed point, then similar areas are automatically identified, and replaced with the customized visual channels, as shown in Figure 4.

There are also many tools for slice illustration. Users can use the brush tool to simply do some drawing on the slice, or use the eraser tool to erase the previous illustration results with original background slice data or cancel the previous iteration steps. We provide a rainbow tool for users to choose different colors to apply to the brush, or directly apply to the selected region. Transfer function is also a frequently-used method to apply color and transparency scheme onto the volume data. Moreover, we use a sketch-based flood-fill method inspired by Liu et al.[2]. This method allows users to customize filling symbols to fill in the

selected area. For example, hospital experts can use blood vessel textures to represent blood vessels. In petroleum exploration, users can use swamp textures to fill swamp geolocation areas.

In the slice analyzer, there are two views for each client. The right view shows the current illustration result, and the left one shows the original slice for comparison. Experts can explore the result more clearly by comparing the left and the right views. With these various selection and coloring tools, experts are able to conveniently edit slices and create a better illustration result.

Algorithm 2 `slicing_floodfill()` function.

```
function SLICING_FLOODFILL(img, current_point, current_mode, threshold)
    queue.push(current_point)
    if current_mode==mode1 then
        pixel = current_point
    end if
    while queue is not empty do
        p = queue.pop()
        if current_mode==mode2 then
            pixel = p
        end if
        for i ∈ {-1,1} do
            for j ∈ {-1,1} do
                current_point = pixel + Point(i, j)
                if abs(img.at(current_point) - pixel) ≤ threshold and img.at(current_point).tag==false then
                    queue.push(current_point)
                    img.at(current_point).tag==true
                end if
            end for
        end for
    end while
end function
```

3.3 | Slice and Volume Integration

With the slice analyzer, the client will send the illustrated slice back to the server in real-time. The server summarizes all the illustrated slices and attaches them to the corresponding position in the original volume to achieve an overall hybrid 2-D and 3-D illustration result, as shown in Figure 5. To visualize slice and volume integration, we choose the ray-casting algorithm mentioned in 3.1, rather than simple texture mapping for slice visualization in cube data. Comparing with the traditional texture mapping method, it could integrate data in different dimension (2-D and 3-D) more clearly. We strings all of the slices together to achieve the 3-D exploration effect. This integration process helps domain experts to fully share their data, which further improves the efficiency.

4 | RESULT

In the experiment, we use a workstation and two low-end laptop computers. The workstation is equipped with Intel Core i7-6700 CPUs operating at 2.70GHz, 16 GB RAM and NVIDIA 1080Ti. The laptop is equipped with Intel Core i5-6700 CPUs operating at 2.70GHz, 4 GB RAM and AMD R9M375. The datasets we used are three CT scanning data and one seismic wave reflection data, provided by experts in different domains.

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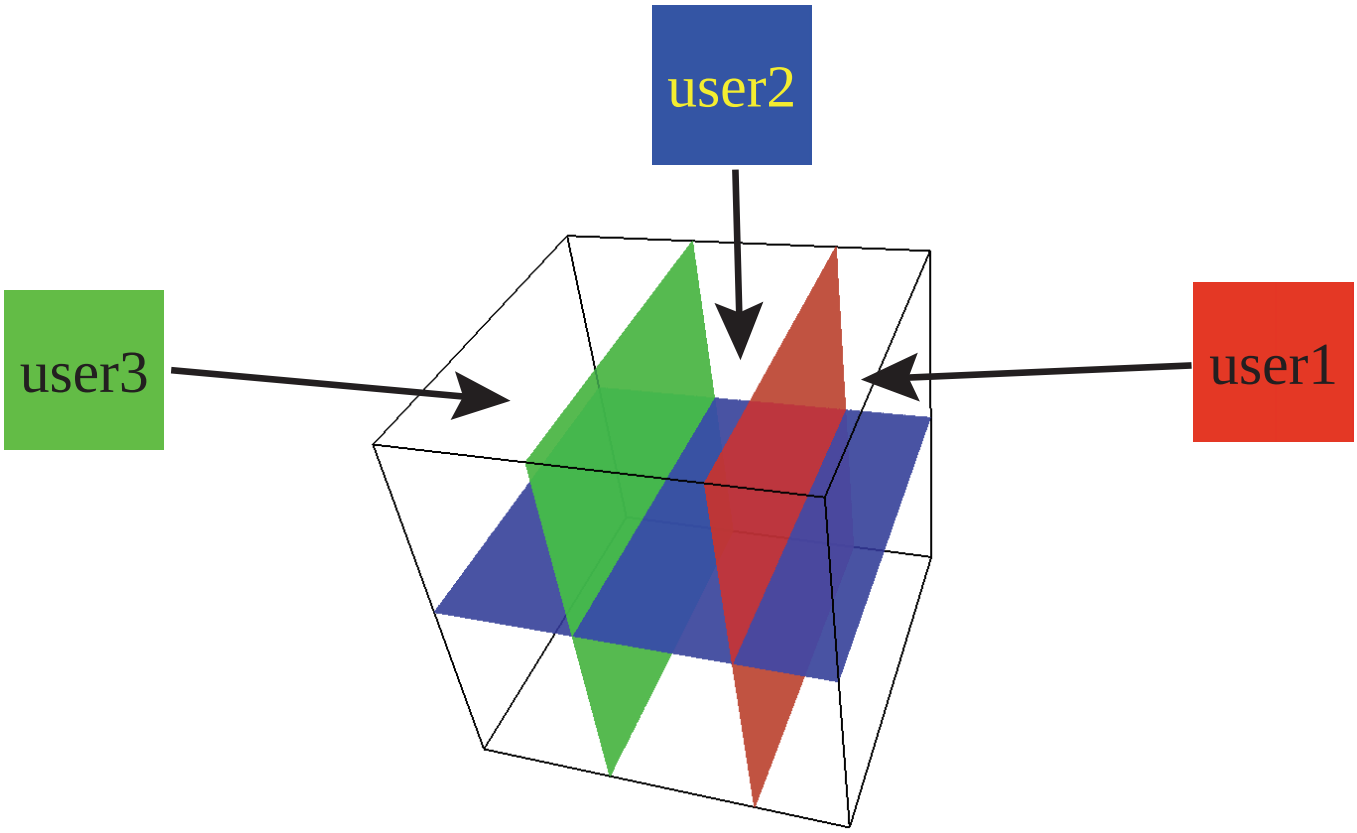

FIGURE 5 Textures for cubes. In the ray interaction, each user processes his/her own slice, which was marked with different colors on the way.

4.1 | Data Description

Dataset	Size	Slice Thickness	Source
Hand	257×124×244	-	CT
Chest	120×192×192	-	CT
Head	256^3	0.02×0.02×0.015	CT
Seismic	21×251×901	2ms	Ultrasonic

TABLE 1 Description of the test datasets.

The first data is a CT scanning data for human hand with the resolution $257 \times 124 \times 244$. The second one is a CT scanning data for human chest with $120 \times 192 \times 192$. The third one is a CT scanning data for human head with $256 \times 256 \times 256$. The last one is a seismic reflection volume data with $21 \times 257 \times 901$, the reflection time interval between two adjacent slices of which is 2 milliseconds.

4.2 | Results of Collaborative Volume Visualization

In the multi-user collaboration system, we allow each user to apply for a specific boundary range in 3-D volume dataset. Each client is allowed to select its boundary range for a sub-volume, which will be rendered on the client locally if the client is available. The server will render all the volume data for all clients. If the selected sub-volume is too large to handle for a client, the server will send the rendered image to the client in real-time. Each user has his own local transfer function. The local transfer function

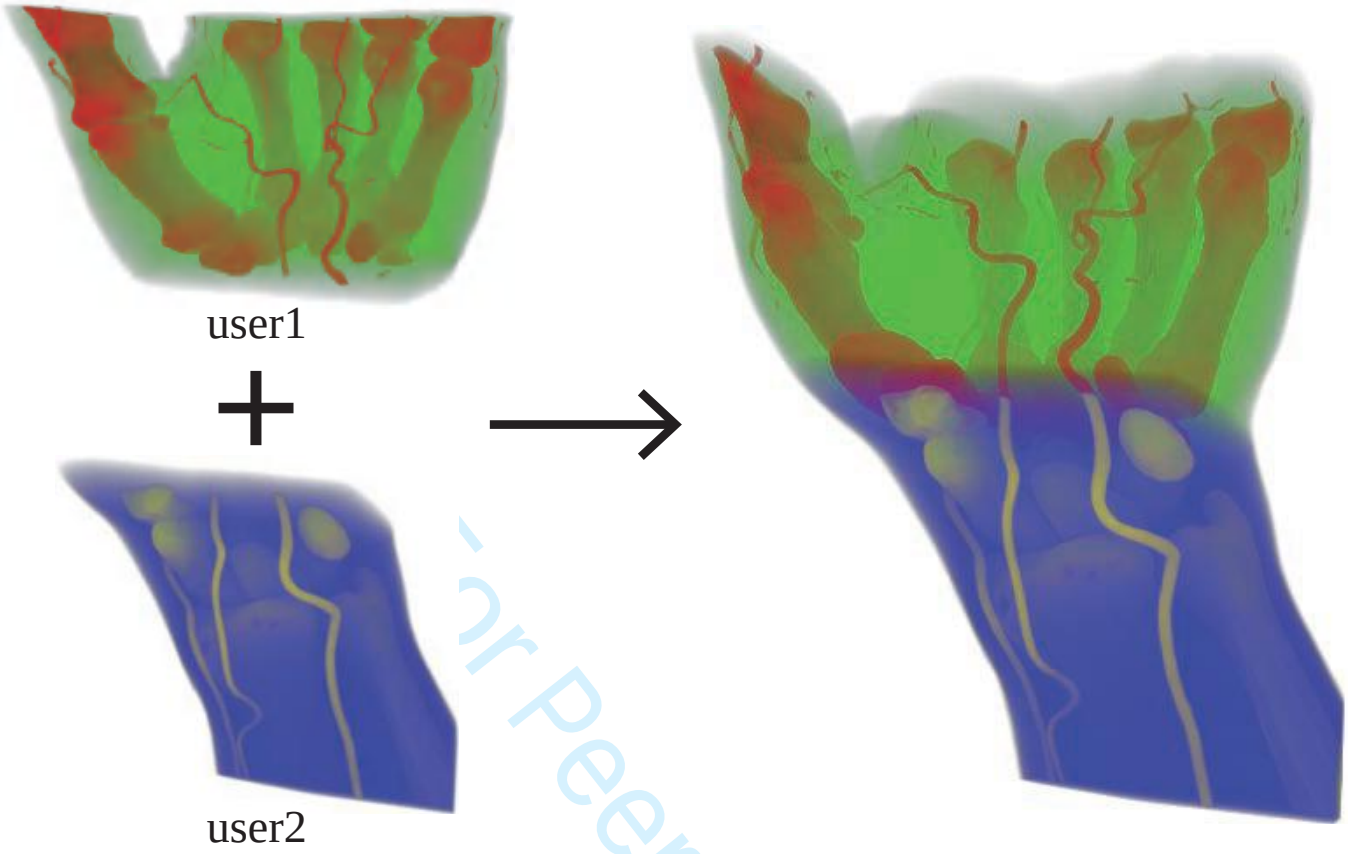

FIGURE 6 Multi-user collaboratively illustrate the dataset hand. The two users conduct collaborative explorations on the blood vessels and the bone.

could be exploited for each user to edit the 3-D volume data, which can provide users with different color and transparency schemes to highlight the region-of-interest.

When users want to edit the 3-D volume visualization result on the server, they must specify the boundary ranges of the sub-volume which they want to illustrate. For example, when a user wants to edit the upper half of the hand data in Figure 6, they need to specify their own range values. After that, the system will judge whether it conflicts with the geometrical boundary of other users. For example, Figure 6 shows the boundary range application example. User1 in applies for the palm area of hand data, user2 can only apply for other areas.

The final visualization result will be integrated and displayed in real-time on the server side because the rendering for the whole data is often computation-intensive. The overall illustration result will rendered on the server side. Any changes performed on the data from any clients would be updated on the server in real-time.

In some cases, the volume data are often some of large 3-D dataset, which are too large to handle on the client. Thus the volume rendering for the whole data can be transferred to the server. The clients are expected to illustrate the sub-volumes, which may match the domain background of the users. For example, the dataset lung in the medical field also contains data such as tissues, blood vessels and bones in the chest, they can assigned to pulmonologists, cardiologist and orthopedist to illustrate. Two pulmonologists like user1 and user2, select the left lung and right lung regions individually for collaborative illustrations, as shown in Figure 7. Furthermore, when they analyze the corresponding lesions, they can also explore the real-time mutual illustrations on the server. Similarly, five users are also allowed to illustrate different sub-domains and slices for the seismic data, as shown in Figure 8.

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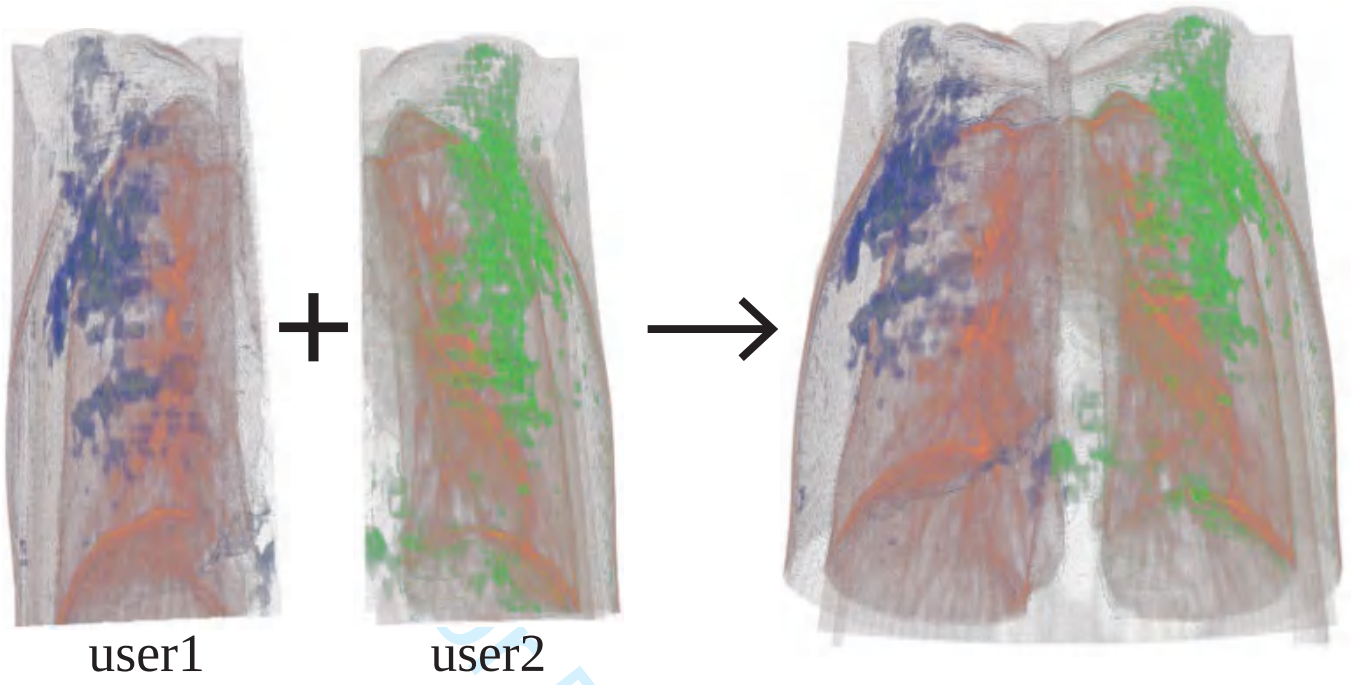

FIGURE 7 Multi-user collaboratively illustrate the dataset lung. Two users explored the left and right lungs to find the lesions, of which purple and green represent the lesions.

4.3 | Results of the Slice Analyzer

The slice analyzer provides some sketch-based interactions, e.g., lasso selection, region-of-interest illustration, local transfer function for the regions-of-interest, texture definition, flood-fill based texture filling, and some basic sketching tools such as pencil and eraser, as the results shown in Figure 9.

Specifically, users are allowed to illustrate the slice through local transfer function followed the design in volume visualization. In Figure 9(b) and Figure 9(f), users can achieve the visualization of the head data and the hand data by local transfer function, respectively. The interaction scope of local transfer function is just restricted in the selected region, as shown in Figure 9(c) and Figure 9(g). With the local transfer function editor, different organs and tissues in the head data are extracted, and the soft tissue, which is difficult to distinguish, is revealed clearly in the hand data.

If one user is interested in the brain tissue of the head slice data, as shown in Figure 9(c), he/she can use the lasso to select a brain tissue, and then use sketch-based illustration tool (e.g., lasso tool, local transfer function editor, pen, eraser, etc.) to illustrate the area. Similarly, if a user is interested in the tendon tissue of the hand slice data, as shown in Figure 9(g), he/she also can use the sketch-based tools to illustrate the tissue.

For the region which contains a plenty of different tissues, seed point tracing algorithm can be used to further illustrate the slice, the illustration results are shown in Figure 9(d) and Figure 9(h) for the head data and hand data, respectively. One of the users is interested in the cerebral cortex area of the brain slice data and the soft tissue of the hand slice data, but these two parts are complex and difficult to recognize. In order to edit the cerebral cortex area and the soft tissue area of the hand efficiently, we can select the accuracy of seed point recognition and then clicks on the area to release the seed point. The nearby area within the threshold range can be replaced with colors for quick and accurate editing.

For the regions which users wants to highlight particularly, they can use the pencil tool or the seed point selection tool to specify certain color or texture to the selected area. Complex brain slices can be illustrated more concisely through texture replacement and seed point editing. Furthermore, the key parts of the soft tissue slice visualization result which are otherwise difficult to discern can be illustrated more clearly, so that even non-professionals can understand it as well, as shown in Figure 9(d) and Figure 9(h).

For the seismic data, users are also allowed to select sub-regions in the slice analyzer. For example, in Figure 10(a), the noisy seismic data slice contains various geological structures and materials. The domain experts want to illustrate and visualize the

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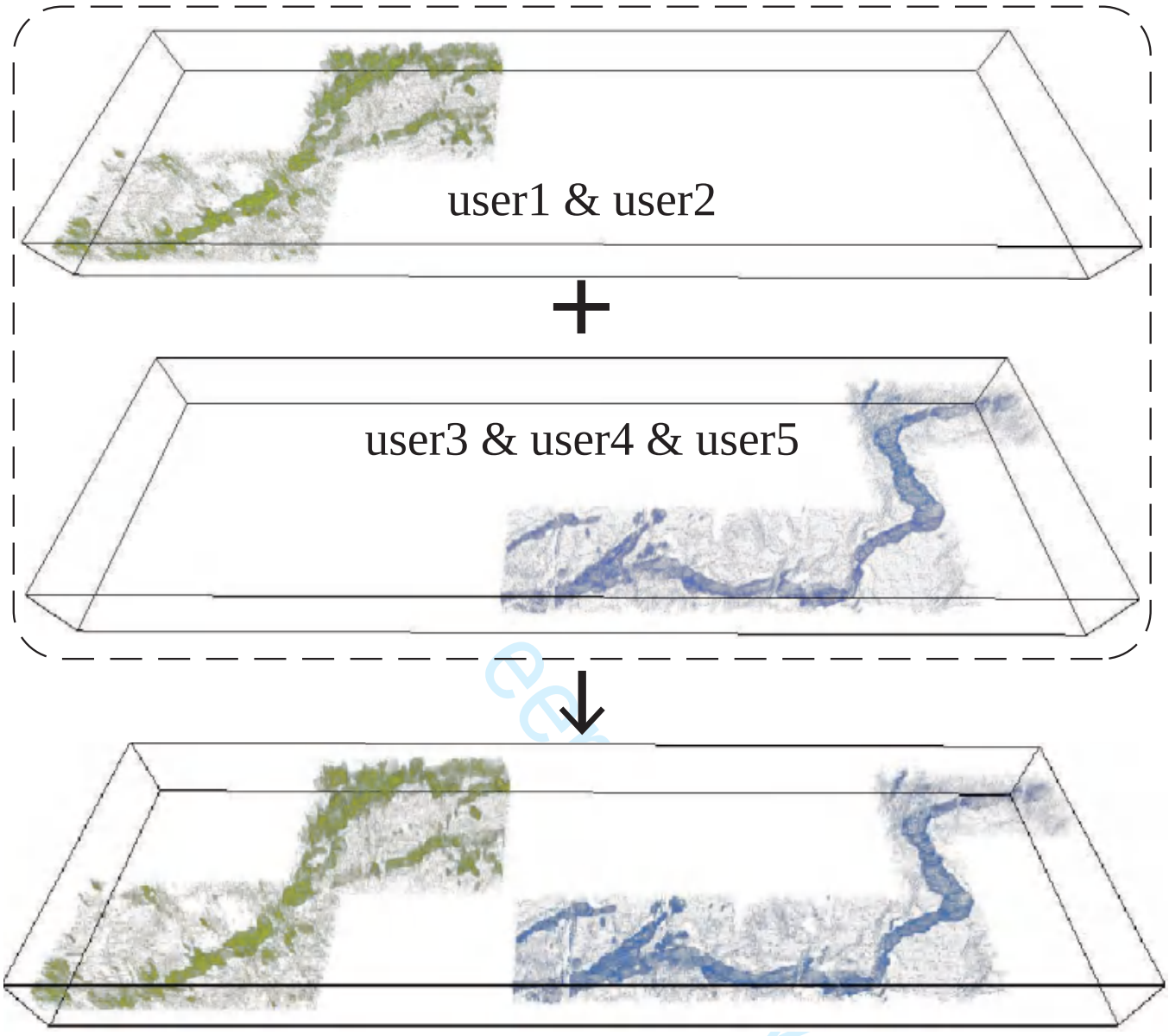

FIGURE 8 The illustration of five users exploring five different sub-volumes of from the seismic data to find the underground flow path. They found two underground path flows which are yellow and blue.

underground flow path in the slice data. They can use the lasso tool to select the approximate area of the underground flow, and then use the local transfer function editor to get a clear underground flow path, as shown in Figure 10(b) and Figure 10(c). They can use the rubber tool to erase other uncorrelated areas, and get the final illustration of the underground flow paths, as shown in Figure 10(d).

4.4 | The Fusion of Collaborative Volume Visualization and Slice Illustration

The illustrated slices can be placed into the corresponding positions of the 3-D volume, which follows a focus+contexts exploration scheme. The illustrated slices provide the detailed information about the focus region-of-interest, while the volume visualization provides the context information.

The system also allows domain experts to explore illustrated slices on the server after they finish analyzing their own slices. The sub-volumes together with the illustrated slices from multiple users will be displayed simultaneously on the server. One

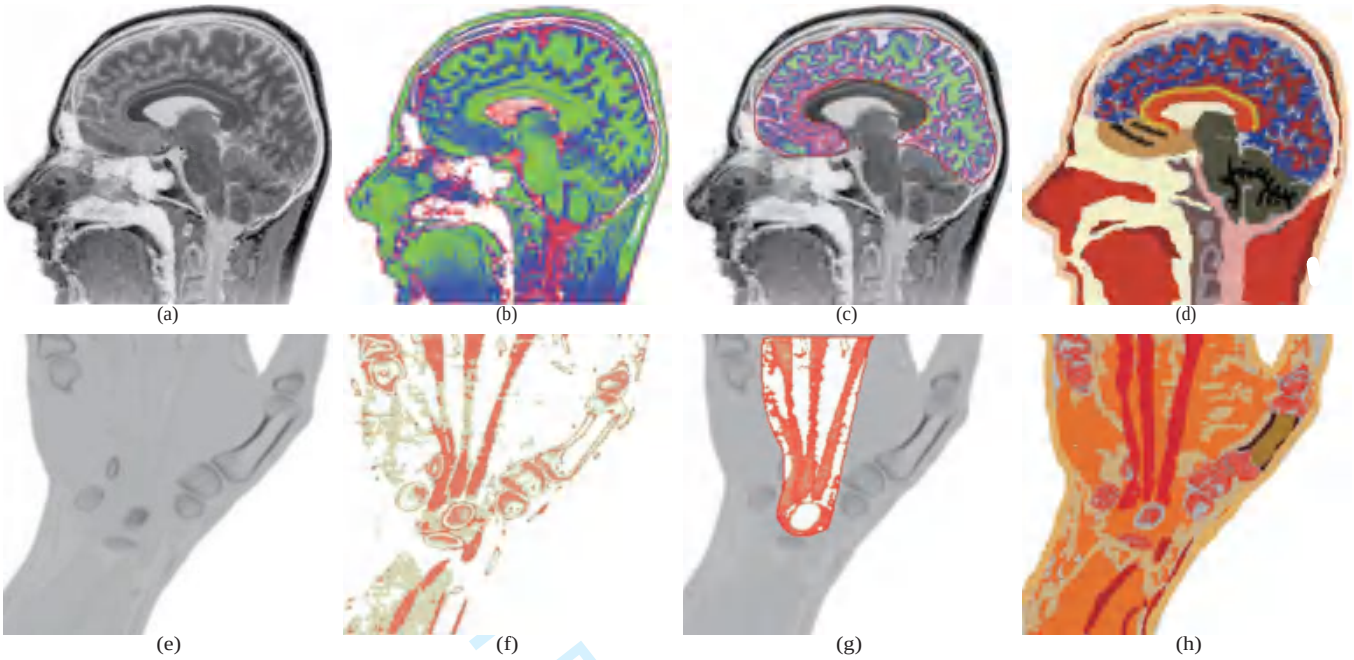

FIGURE 9 Medical slice data illustrated by the user of a client. (a) A 2-D slice selected by the user from the dataset head. (b) Slice illustration by local transfer function. (c) Regions-of-interest selected by lasso tool for the dataset head. (d) Use seed point tracing and lasso to illustrate the slice. (e) A 2-D slice selected by the user from the dataset hand. (d) Slice illustration by local transfer function. (f) Regions-of-interest selected by lasso tool for the dataset hand. (h) Use seed point tracing and lasso to illustrate the slice.

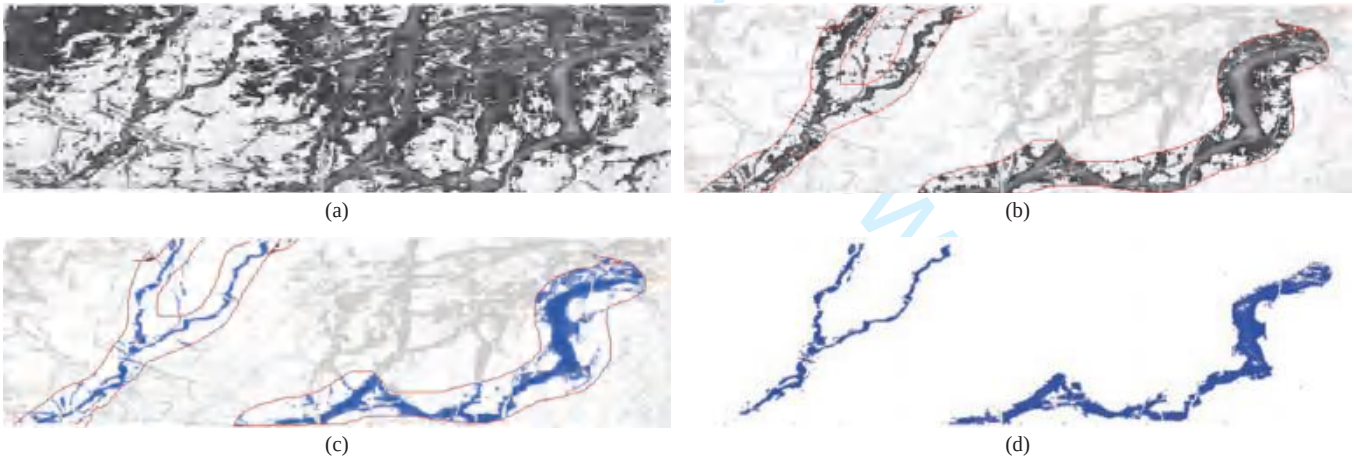

FIGURE 10 Seismic slice illustrated by the user of a client. (a) A 2-D raw seismic slice selected by the user. (b) Regions-of-interest selected by lasso tool. (c) Regions-of-interest illustrated by local transfer function. (d) The underground flow path is extracted and illustrated by a series of illustration operations.

user can observe the real-time 2-D slicing visualization result analyzed by other users on the server. For example, user1 edits the upper half of the hand visualization result, and user2 edits the lower half of the hand visualization result and the slices in the 3-D volume dataset, all the illustrated results will be fused into a view and can be explored on the server, as shown in Figure 11.

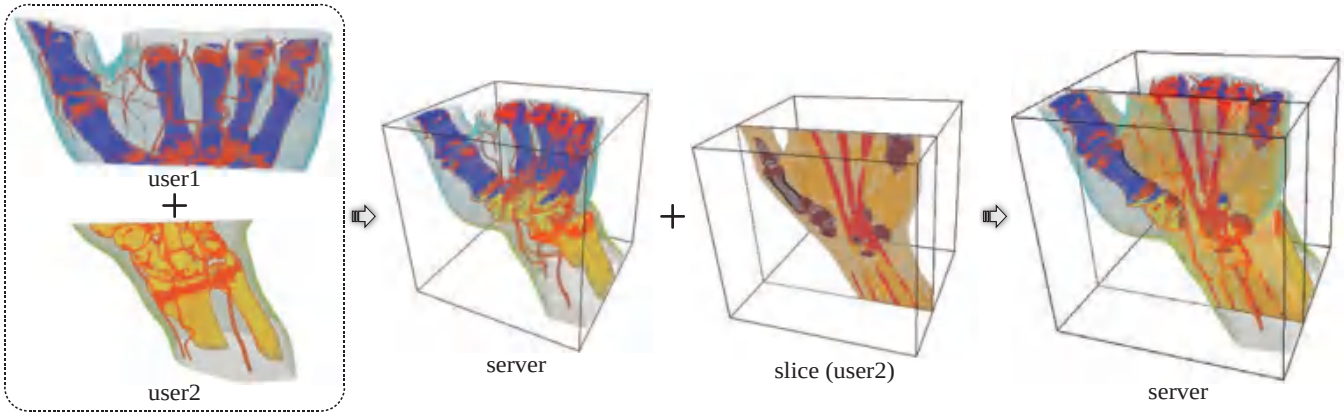

FIGURE 11 Multi-user collaborative illustration for the dataset hand. User1 and user2 illustrate their corresponding sub-volumes and slices. All the 3-D and 2-D illustrations can be achieved from all the clients and are fused on the server.

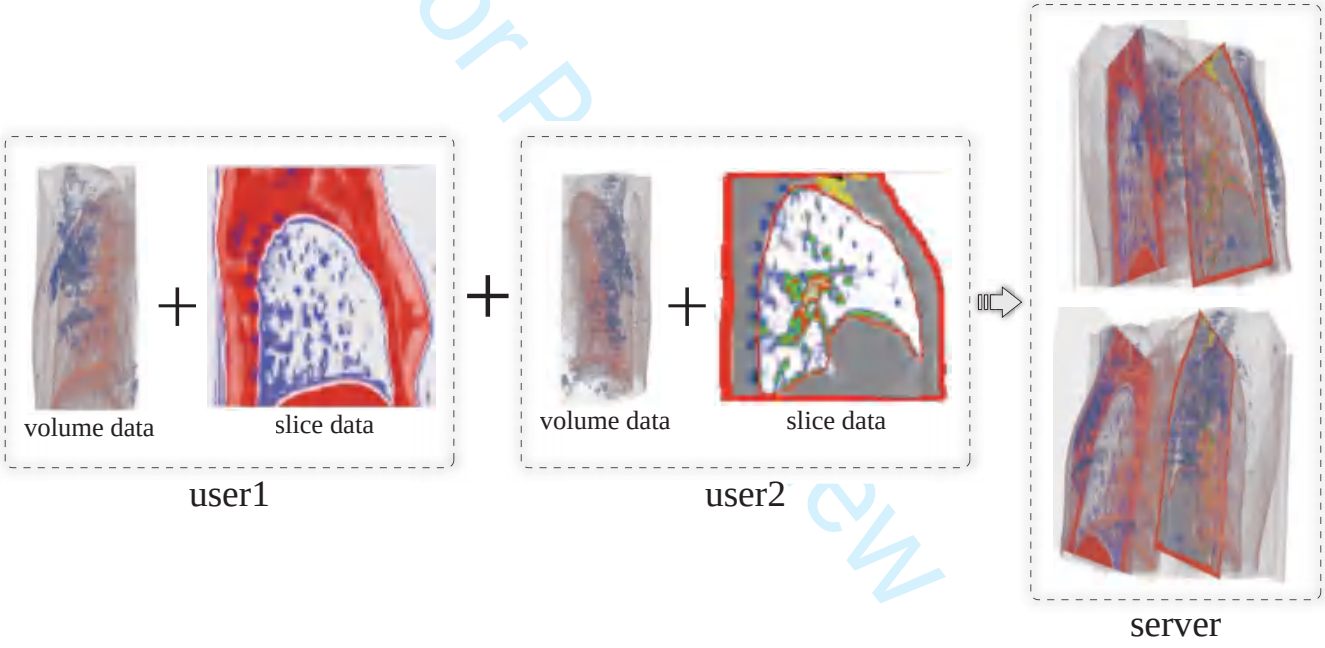

FIGURE 12 The multi-user collaborative illustration for the dataset lung. User1 and user2 are allowed to illustrate their corresponding sub-volumes and slices. All the 3-D and 2-D illustrations can be achieved from all the clients and are fused on the server.

Similarly, two users are allowed to edit the volume visualization result and slices, the final fused result will be rendered on the server side, as show in Figure 12. Additionally, with the help of the slice analyzer, users can explore the soft tissues that are difficult to observe in 3-D volume dataset, such as the tendon tissue of the HAND and the organized system of LUNG.

4.5 | Evaluation

We conducted serval experiments and evaluate the results by consulting different medical domain experts who have been engaging in CT/MRI data imaging and analysis for many years. Then we find some cases to show the usability and effectiveness of the proposed approach.

4.5.1 | Expert Feedback

We received many valuable feedback on the system. The domain experts are doctors in different departments. The major feedback is listed as follow:

- “The multi-user collaboration system allows multiple users to extract the tissues they want to see at the same time, which is a great help in the peer-expert discussion."
- “The slice analyzer can efficiently and quickly illustrate the slices, which can be a good application in medicine teaching."
- “The multi-user collaboration system is expected to provide a good solution in simulated surgery for multiple experts."
- “The current 3-D visualization system we used is limited to orthopedics and angiography, all of which are single colors. Only salient bones or blood vessels can be seen. Instead, the soft tissues can be clearly identified by using the proposed approach. The information is significant when it comes to liver tumors or other operations."
- “For the analysis of tumor data, there is still a lack of tissue recognition, if the tool can recognize most of the tissues and tumors, it will be quite helpful in tumor surgery, because it will help us clearly see the tumor size in surgical planning."

Most of the feedback is positive. Medical experts believe that the proposed approach can help them quickly find the position of pathological tissue, which is of vital importance to analyzing the same case. In addition to allow experts to illustrate CT/MRI images in more detail, they think the slice analyzer could also make their medicine teaching more intuitive. With the designed slice analyzer, they can illustrate the patients’ condition better and thoroughly, rather than merely using monochrome CT slices to illustrate cases.

4.5.2 | Usability

We also evaluate the usability of the proposed system by asking four users ($tf1 - tf4$) to participate the study. $tf1$ is trained to be familiar with the system, while $tf2$ and $tf3$ having basic knowledge of computer science, and $tf4$ being a non-computer professional. We make a simple and fast training for the users about the usage of the system. The first task was to find the corresponding tissue or position in the dataset through transfer function editor, respectively. The participants were asked to illustrate the data by the volume explorer and the 2-D slice analyzer. The participants processed an identical data collaboratively, each of them focus on a subset of the dataset, where they were asked to select the appropriate tissue or fragment. We received many feedback from the participates. Most of the feedback from the users is positive, which demonstrate the usability of the proposed approach.

Users	Dataset #01	Dataset #02	Dataset #03
$tf1$	5.3	6.5	11.3
$tf2$	15.6	20.4	15.6
$tf3$	16.3	21.5	16.8
$tf4$	21.4	25.6	26.7

TABLE 2 Performance test (in seconds) for the datasets HAND(Dataset #01), HEAD(Dataset #02) and SEISMIC(Dataset #03). The performance values refer to the average number of the time it takes to achieve the desired effect when illustrating different data.

Table 2 shows the average timing performance comparison between different participants when they conducting queries by using the 2-D slice analyzer to illustrate the three data. From the four groups of test results, we can find that different users have much different time performance in data processing, hence, we can make a conclusion that it requires some training before usage. The reason why $tf4$ takes about more time is that the interactive tissue select and cut are time consuming, and $tf4$ is not familiar with the volume data illustration.

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4.5.3 | Case Study

In our performance test, the proposed system performs better in ïňĄne-tuning than the traditional single-user single-machine systems. It is hard for a single user to ïňĄnd the concretes of some tissues or positions of large volume datasets. The corresponding timing results consumed by the traditional single-user single-machine systems and the proposed system are shown in Table 3 and Table 4.

Test group	Test group #01	Test group #02
Single user system	10.7	11.6
Multi-user system	8.9	9.6

TABLE 3 Performance test for the dataset LUNG. The performance values refer to the average number of the time it takes a user to illustrate lung data in a single person and the time it takes multiple people to illustrate lung data at the same time.

Test group	Test group #01	Test group #02
Single user system	15.6	13.5
Multi-user system	10.8	9.6

TABLE 4 Performance test for the dataset SEISMIC. The performance values refer to the average number of the time it takes a user to illustrate seismic data in a single person and the time it takes multiple people to illustrate seismic data at the same time.

We find that the consuming time of our system is obviously less than the traditional single-user system on the whole, as shown in Table 3 and Table 4. From the Table 4, using traditional single-user system, the average time of five users is 14.1s. The average time of the proposed system is 10.2s. The time of the traditional system is about 1.4 times of the proposed system. Additionally, we can find it takes much more time to conduct selecting tissues or positions on dataset SEISMIC than dataset LUNG. The reason is that the number and complexity of the data in dataset SEISMIC is obviously larger than that of dataset LUNG.

5 | DISCUSSION

The system provides a multi-user collaborative analysis for volume data illustration and slice data illustration. However, the proposed system still has many limitations:

First, the system needs users to get involved in the illustration by using their background knowledge. For the slice analyzer, it needs to specify the sub-volume boundary for each users before his/her illustration. In the slice transfer function editing, users need to pick up a foreground pixel as a seed to generate a Gaussian function to obtain the intensity range. If the picked pixel can be used to well distinguish the foreground tissue from the background tissues, it will greatly reduce the illustration time.

Second, it would be more helpful if the system can provide more various interactions between the slice and the volume data set. For example, the domain experts are expected to just select a seed point in the slice, which would trace the whole 3-D object the seed point locate in the 3-D space.

Third, the slice analyzer can often achieve good illustration results for CT/MRI data, but the 3-D illustration results seem to be not good enough when it comes to noisy and discontinuous seismic data. We plan to design a more robust algorithm to conduct collaboration on seismic data.

6 | CONCLUSION

In this paper, we propose a multi-user collaborative interaction visualization system. The system overcomes the limitation of single user interaction in volume data visualization. It enables everyone in the expert group to work according to his own field and complete the exploration of 3-D data collaboratively.

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The 2-D slice analyzer provides many interactive tools to explain slices, such as pencil, eraser, transfer function, etc. They can help users to better explain slices. In order to better observe the effect of the slice, the user can paste the interpreted slice into the corresponding position in the volume data by texture. Finally, the volume data edited by each user and the edited slice are merged, and the effect of 3-D display allows the user to observe more clearly.

Though the proposed system has some limitations on automatic illustration, so we plan to continue our work to improve the system. First, we plan to develop some new features for multiple users, especially in adjusting the range of the volume data and the detailed visualization for a single user in the collaborative environment. We will develop more intelligent tools to improve user experience. Second, the interpretation of slices is still a time-consuming problem, so we plan to employ machine learning method for the analysis of slices. Third, the fusion of multiple volume data and slices needs further optimization to improve the rendering speed of the machine, so that low-end graphics cards will have less pressure running it.

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